

Two-Anchor Holdout/Hermite: Solving the TII-254 McEliece Key Recovery Challenge

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Abstract. We report on the solution to the TII-254 McEliece key recovery challenge – currently the hardest solved challenge under the original brute-force metric (2^{254}). The parameters of TII-254 are $(m, t, n) = (8, 12, 223)$, defining a binary $[223, 127]$ code specified by a full-rank (96×223) parity-check matrix. We state the method as a thirteen-step process, separating heuristic and non-heuristic choices. At a high level, we computed two complete 121-dimensional relation kernels conditioned at distinct public coordinates, combined them to isolate a certified 80-dimensional pair core, removed a 64-dimensional common nuisance space, and identified the remaining 16 dimensions as an $\mathbb{F}_2$ s projective-line geometry. This yielded all 87 visible locators, after which a deterministic completion search recovered the full support and polynomial. The two final Krylov sequences alone used 27.2 GPU-hours on NVIDIA GH200s, excluding GPU reconstruction and CPU processing. We provide a self-contained artifact with compact recovery inputs and code, an independent key verifier, and Lean proofs of the reusable linear-algebraic steps.

Keywords: McEliece · Binary Goppa Codes · Key Recovery · Holdout attack · Block Wiedemann

1 Introduction

An error-correcting code adds redundancy to data so that a receiver can repair a limited number of errors. In 1978, McEliece turned this idea into one of the first public-key encryption schemes: a public binary matrix lets the sender form a codeword and deliberately add errors, while a hidden algebraic description lets the receiver remove them efficiently [9]. An attacker sees only the matrix and faces the much harder problem of decoding an apparently unstructured code.

The original formulation of the McEliece cryptosystem survived more than four decades of cryptanalysis remarkably well, eventually leading to its standardization by ISO in 2026 [6]. However, 2026 also brought a flurry of cryptanalytic results – our method builds most directly on recent work by Ghoshal et al. [3], Vedenov [10], and Hemmert [5]. These are *structural key-recovery attacks* that reconstruct the hidden algebraic description (equivalent to a secret key), allowing ciphertexts to be read as easily as by the legitimate recipient.

The TII McEliece Challenges. The Technology Innovation Institute (TII) in the UAE introduced public McEliece challenges in 2023 to measure progress in cryptanalysis.* We will focus on its algebraic key recovery track in this work. The numerical labels of the challenges were “brute force” work-factor estimates, not cryptographic parameters. Successive attacks have shown that the labels should not be read as current security levels: Bardet, Lemoine, and Tillich recovered instances through TII-210 [1], while Ghoshal et al. developed the Holdout/Hermite construction used here [3]. Ghoshal et al. report that Steve Weis, assisted by Anthropic’s Claude, combined their approach with related ideas to recover TII-253, the highest-numbered solved TII challenge at the time [3]. Weis announced the result on 19 August 2026.** This paper describes our recovery of TII-254 on September 8, 2026, together with the methods we used.

Outline of our TII-254 attack path. The central “theme” of the attack is elimination by redundancy: We first derive a very large homogeneous linear system from degree-six products of public equations. Most solutions of that system are predictable algebraic identities that contain no information about the hidden key. We prove and remove 4,617,031 such *nuisance directions*. We then solve two related versions of the remaining system, each requiring relations to vanish at a different selected public observer (the anchor). Neither solution space is useful by itself. Their overlap, however, lets us cancel the remaining nuisance structure and expose a 16-dimensional binary space behaving as a two-dimensional vector space over $\mathbb{F}_{256}$. Its one-dimensional $\mathbb{F}_{256}$ subspaces are the hidden projective locators.

Our main contributions:

1. We present a 13-step end-to-end key-recovery process that starts with the McEliece public key matrix and ends with fresh full-key verification.
2. We describe three structural reductions and exact certificates that allowed the target computation to proceed decisively: coordinate deflation, complete-kernel extraction, and relative completion of the second anchor.
3. We introduce a two-anchor pair core technique that we used to isolate a direct $64 + 16$ decomposition for TII-254; this avoided a potentially exponential unprojection stage.
4. We describe the AI-driven algorithm improvement, the CUDA/GPU implementation cycle, and the measured GPU and CPU resources used in the successful recovery.
5. The accompanying artifact contains compact recovery inputs and code, a standalone independent Sage verifier, CUDA and CPU sources for the large linear-algebra stages, and Lean-checked versions of the reusable theorems.

* TII McEliece Challenge: <https://crowdchallenge.tii.ae/mceliece-challenges/>
Challenge data: https://github.com/ElenaKirshanova/tii_decoding_challenge

** Steve Weis, post on X, 19 August 2026, <https://x.com/sweis/status/2089904907054903792>.

2 Background and notation

Statement terminology. We distinguish three kinds of statements throughout. A *theorem* holds for every input satisfying its stated hypotheses. In our context, an *exact certificate* is a deterministic computation on this particular public instance and can be replayed. A *heuristic* motivates a choice such as a projection or an anchor (but would not be used to, e.g., accept a recovered key.)

The TII-254 key recovery problem. Let $\mathbb{F}_2$ be the binary field and let $F = \mathbb{F}_{2^m}$. For TII-254, $m = 8$, the support length is $n = 223$, and the Goppa polynomial has degree $t = 12$. A binary Goppa decoding key consists of distinct support elements $\alpha_1, \dots, \alpha_n \in F$, one assigned to each code position, and a monic irreducible polynomial $g \in F[X]$ with $g(\alpha_i) \neq 0$. It defines a binary code through the extension-field parity checks

$$\sum_{i=1}^n c_i \frac{\alpha_i^k}{g(\alpha_i)} = 0 \quad \text{in } F, \quad 0 \leq k < t, \quad (1)$$

where $c_i \in \mathbb{F}_2$. Expanding the t equations in an $\mathbb{F}_2$ basis of F gives at most $mt = 96$ binary checks. The challenge publishes a full-rank 96×223 binary parity-check matrix H for this code and hides (α, g) .

Two keys are *equivalent* when they define the same binary code and therefore the same public row space. This is the relevant success condition for key recovery: an equivalent Goppa description supplies a decoder even if it differs from the author's original representation. We make no claim to have recovered the author's exact encoding.

The Holdout/Hermite Construction. Our approach mainly builds on the framework of Ghoshal et al. [3]. The geometric recovery viewpoint draws on Vedenov's derivative-space approach [10]. The method shares the use of higher-order polynomial relations to expose hidden code structure with Hemmert's higher-order-vanishing key recovery [5].

The degree-six Holdout/Hermite construction used here begins by choosing k independent public columns, called an information set, and using them as a normalized coordinate system for the remaining columns. For TII-254, $k = 50$. A Hasse derivative of order r is the coefficient of Z^r in $f(X + Z)$; the collection of a value and such derivative data at a point is its Hasse jet. Degree- d square-free monomials in the public coordinates form vector spaces $\mathcal{A}_d$ with bases indexed by d -subsets of $\{0, \dots, 49\}$. Multiplication by public linear or quadratic forms and the public Hasse-jet constraints define a sparse relation operator

$$\mathcal{E} : \mathcal{V} \longrightarrow \mathcal{A}_6, \quad (2)$$

where $\mathcal{V}$ is a direct sum of coefficient blocks. The kernel, or nullspace, $\ker \mathcal{E}$ consists of the inputs that $\mathcal{E}$ maps to zero; each such vector is a degree-six public relation. Earlier work explains why a small, geometrically structured part

of these relations can reveal Goppa locators [3,10]; the difficulty is separating it from a much larger space of universal product and cycle identities.

We use 87 ordinary public observers, whose public-column labels form a fixed set $\mathcal{J}$. An observer is a public linear map that records how a relation behaves at one labeled column. Conditioning on an *anchor* means forcing the relation to vanish at one selected observer.

Anchor selection. The anchor candidates are the labels in $\mathcal{J}$, ordered by their original, zero-based public-column indices. This order begins 2, 5, 7, 8, 10, $\dots$; labels 0, 1, 3, 4 are outside the observer set. We take the first entry, $a_1 = \min \mathcal{J} = 2$, and the next distinct entry, $a_2 = \min(\mathcal{J} \setminus \{2\}) = 5$. Thus the label choice uses only membership in the public observer set and its fixed order. Both labels were fixed before the corresponding supplier output was known. The codec and complete-kernel checks in Sections 3.2 and 3.3 determine whether these choices yield usable kernels; the ordering itself supplies no guarantee of successful recovery. Neither choice uses hidden locators.

For anchor a , let J_a inject its compressed coordinate domain into the common physical relation space $\mathcal{V}$, let R_a be the output compression, and write

$$B_a = R_a \mathcal{E} J_a, \quad \hat{K}_a = \ker B_a, \quad K_a = J_a \hat{K}_a. \quad (3)$$

The hat therefore marks a compressed-domain kernel, while K_a is its replay-certified physical lift in the common space $\mathcal{V}$. Intersections such as $K_2 \cap K_5$ always refer to these physical lifts. We reserve A_j for the local kernel of observer j .

All dimensions below are over $\mathbb{F}_2$ unless a different field is stated. For a subspace U , $U^\perp$ is its annihilator under the standard pairing. “Rank” always means exact finite-field rank, not a floating-point estimate.

3 The TII-254 Key-Recovery Process

We present the successful key recovery route as a process, together with theorems and other technical rationale for it. The experiment design is fail-closed: every incomplete rank certificate produces INCONCLUSIVE, never a negative cryptanalytic claim. The numerical values shown are those obtained for TII-254.

Our description separates supplier work from deterministic recovery. Steps 3–8 supply and certify two complete kernels. Steps 9–11 convert them into projective locators. Steps 12–13 are a bounded completion and a definitive public-key test. The following sections offer mathematical justifications for each acceptance gate; detailed proofs and formalization pointers are collected in the appendices and the artifact.

Input: The authenticated 96×223 public binary parity-check matrix H .

Output: A support $\alpha \in \mathbb{F}_{256}^{223}$ and monic irreducible degree-12 polynomial g , or INCONCLUSIVE.

1. **Authenticate and derive.** Rank-check H and derive the public degree-six Hasse-jet relation operator $\mathcal{E}$, its 87 observers, and a minimal dependent set of public columns, called a circuit. (Section 3.1)
2. **Deflate nuisance.** Construct $L_\star \leq \ker \mathcal{E}$ with $\dim L_\star = 4,617,031$ and a triangular coordinate information set; restrict $\mathcal{E}$ to the complementary 15,654,269 coordinates.
3. **Condition anchor 2.** Apply initial public observer-2 conditioning and replayable range compression to obtain $B_2 = R_2 \mathcal{E} J_2$ of dimension $N = 15,527,170$. (Section 3.2)
4. **Generate moments.** Compute 60,718 width-512 block-Krylov moments $U_2^\top B_2^{i+1} Y_2$, with restart inventories and transpose tests.
5. **Extract relations.** Compute and validate a polynomial matrix relation; derive finite-Hankel rank 15,527,049 and hence the upper bound $\dim \hat{K}_2 = \dim \ker B_2 \leq 121$.
6. **Certify K_2 .** Reconstruct a 121-dimensional compressed panel $\hat{Q}_2$, require $B_2 \hat{Q}_2 = 0$ and literal $\mathcal{E}(J_2 \hat{Q}_2) = 0$, conclude $\hat{Q}_2 = \hat{K}_2$, and set $K_2 = J_2 \hat{K}_2$ with $\dim K_2 = 121$.
7. **Seal the overlap.** Use the next label 5 in the fixed public observer order; restrict this observer to physical K_2 , and replay a basis of $I = K_2 \cap K_5$ of dimension $121 - 15 = 106$. (Section 3.3)
8. **Complete anchor 5.** Repeat Steps 3–5 for label 5, reconstruct $\hat{P}_5 \leq \hat{K}_5$, lift $P_5 = J_5 \hat{P}_5 \leq K_5$, and accept completeness only when $\dim(K_2 + P_5) - 121 = 121 - 106 = 15$.
9. **Certify the pair core.** Restrict the complete physical kernels K_2, K_5 to compatible observer images, form all local-kernel pair sums, and certify their intersection P by a dual witness; obtain $\dim P = 80$. (Section 3.4)
10. **Isolate the carrier.** Compute the common invisible part N_0 with $\dim N_0 = 64$ and quotient $C = P/N_0$ with $\dim C = 16$; require 87 local cuts of dimension 8 and every pair sum of dimension 16. (Section 3.5)
11. **Recover locators.** Split C into two 8-spaces. Normalize labels 66, 114, 145 to 0, ∞ , 1, recognize the graph pencil over $\mathbb{F}_{256}$, and recover the eight locator branches related by Frobenius. (Section 3.6)
12. **Complete the key.** For each branch and each choice in a fixed $170 \cdot 169 = 28,730$ pole/root inventory, factor the characteristic-two numerator, apply public row-space and semantic screens, complete missing locators using Kirshanova–May, and stop when a key verifies. (Section 3.7)
13. **Verify and replay.** Require that g is irreducible, build the full Goppa check matrix from (α, g) , require equality with the public row space, reproduce the transcript in a fresh deterministic recovery process, and check the key with the standalone Sage verifier; otherwise return INCONCLUSIVE.

3.1 Steps 1–2: authenticated equations and coordinate deflation

The literal operator $\mathcal{E}$ has 20,271,300 source coordinates, 15,890,700 output coordinates, and 6,223,124,835 nonzero incidences. Direct black-box linear algebra

at that size wastes effort rediscovering identities implied by square-free multiplication. We instead construct $L_\star$ from pointwise kernels, independent pair cycles modulo the pointwise families, and three smaller cross-block families. Grouping the independent components in Table 3 in this order gives

$$\dim L_\star = 113,925 + 4,469,248 + 33,858 = 4,617,031. \quad (4)$$

Every generator is replayed under the original $\mathcal{E}$. A block-triangular minor supplies the same number of distinct pivot coordinates, proving independence without assuming that the listed families exhaust $\ker \mathcal{E}$.

Theorem 1 (Coordinate deflation). *Let $T : V \rightarrow W$ be linear, let $L \leq \ker T$, and suppose coordinate projection onto a set P restricts to an isomorphism $L \rightarrow \mathbb{F}_2^P$. If S_P is the coordinate subspace vanishing on P , then $V = L \oplus S_P$, $T(V) = T(S_P)$, and $\ker(T|_{S_P})$ maps isomorphically to $\ker T/L$.*

The theorem turns the information minor into an exact compiler: delete the 4,617,031 pivot coordinates and retain a complement of dimension 15,654,269. It does not claim that $L_\star$ is the whole nuisance kernel. It claims precisely what the process needs: no quotient relation is lost and no new relation is created. A proof appears in Section B; the same statement is checked in Lean.

3.2 Steps 3–6: conditioning, Krylov moments, and a complete first kernel

Let $\overline{O}_j$ denote the degree-four observer at label j , taken modulo the observer image of $L_\star$. Its target quotient has dimension 127,163. Imposing $\overline{O}_a = 0$ on the coordinate complement leaves $15,654,269 - 127,163 = 15,527,106$ coordinates. The production codec then adjoins a fixed, explicitly known 64-dimensional subspace $T \leq L_\star$ spanned by independent literal pair relations. Thus J_a parametrizes the direct sum of this conditioned complement and T , giving source dimension $N = 15,527,106 + 64 = 15,527,170$. Every vector of T vanishes under $\mathcal{E}$ and all quotient observers; the same T is retained for both anchors. A systematic Toeplitz map R_a compresses the 15,890,700 outputs to N coordinates and gives the square black-box map

$$B_a = R_a \mathcal{E} J_a : \mathbb{F}_2^N \longrightarrow \mathbb{F}_2^N, \quad N = 15,527,170. \quad (5)$$

The retained zero directions imply $\text{rank}(\mathcal{E} J_a) \leq N - 64$. For a uniformly sampled systematic Toeplitz compressor, this rank gap gives a range-compression failure bound below 2^{-64} by a union bound over the nonzero image vectors. Thus the codec’s +64 implements a compression margin by retaining known kernel directions; it is not extra unknown key information. Range compression is not trusted as an equation certificate: every accepted $q \in \hat{K}_a = \ker B_a$ must also pass the literal test $\mathcal{E}(J_a q) = 0$ before $J_a q$ is admitted to the physical kernel K_a .

Block Wiedemann replaces an explicit $N \times N$ elimination by repeated applications of B_a and $B_a^\top$ [2]. With binary $N \times 512$ starting panels Y_a, U_a , we compute

the 512×512 moments $S_i = U_a^\top B_a^{i+1} Y_a$. A polynomial-matrix minimal-basis computation then gives relations whose coefficients reconstruct candidate kernel vectors [4,7]. The request length

$$2 \left\lceil \frac{N}{512} \right\rceil + 64 = 60,718 \quad (6)$$

provides the finite-Hankel window with 64 extra sequence terms. This independently chosen length margin is distinct from the 64 retained kernel directions and their range-compression role.

Theorem 2 (Shifted-Hankel completeness gate). *Let B be an endomorphism of an N -dimensional space and let $\mathcal{H}$ be any finite block Hankel matrix with blocks $U^\top B^{i+j+1} Y$. Then $\text{rank } \mathcal{H} \leq \text{rank } B$, so $\dim \ker B \leq N - \text{rank } \mathcal{H}$. If a replayed subspace $Q \leq \ker B$ has dimension $N - \text{rank } \mathcal{H}$, then $Q = \ker B$.*

For both anchors, the validated relation data give $\text{rank } \mathcal{H} = 15,527,049$, hence $\dim \hat{K}_a = \dim \ker B_a \leq 121$. The reconstructed label-2 compressed panel $\hat{Q}_2$ has rank 121 and passes both $B_2 \hat{Q}_2 = 0$ and $\mathcal{E}(J_2 \hat{Q}_2) = 0$. Theorem 2 therefore proves $\hat{Q}_2 = \hat{K}_2$; its replay-certified physical lift $K_2 = J_2 \hat{K}_2 \leq \ker \mathcal{E}$ is complete and has dimension 121. The proof of the rank inequality is a one-line factorization through B_a ; formal details are in Section B and the Lean artifact.

3.3 Steps 7–8: relative completion of the second anchor

Computing a second complete kernel from scratch would require the finite relation panel to hit all 121 of its dimensions. The already certified physical kernel K_2 enables a stronger and cheaper test. Label 5 was already fixed by the public ordering when its observer was restricted to K_2 . The measured rank 15 supplies the overlap certificate for this selected pair, giving

$$\dim(K_2 \cap K_5) = 121 - 15 = 106. \quad (7)$$

We explicitly construct this physical 106-space and replay all of its vectors through $\mathcal{E}$ and the anchor-5 condition before seeing the second reconstruction result.

Theorem 3 (Relative anchor completion). *Let K_2, K_5 be subspaces, let $I = K_2 \cap K_5$, and let $P_5 \leq K_5$. Suppose $\dim K_5 \leq \kappa_5$. If*

$$\dim(K_2 + P_5) - \dim K_2 = \kappa_5 - \dim I, \quad (8)$$

then $\dim K_5 = \kappa_5$ and $K_5 = I + P_5$.

For TII-254, the second Hankel certificate gives $\kappa_5 = 121$, while reconstruction supplies exactly 15 new directions modulo K_2 . Thus equality in Eq. (8) certifies the complete K_5 even if the reconstructed panel omits directions already supplied by the 106-dimensional overlap. This deterministic theorem is one reason the second run did not rely on an unquantified “good panel” assumption.

3.4 Step 9: the two-anchor pair core

The observer restrictions require compatibility between the two complete physical kernels. For each of the 85 labels j other than anchors 2, 5, retain $v \in K_a$ only if $\overline{O}_j(v) \in \overline{O}_j(K_b)$, where b is the other anchor. In other words, its observer value must also be attainable from the other kernel; the matching vector may depend on j . Intersecting these linear preimage conditions imposes seven independent constraints on each 121-space, leaving candidate spaces D_2, D_5 of dimension $121 - 7 = 114$. Both contain $I = K_2 \cap K_5$, and their intersection is exactly I . Their physical sum $S = D_2 + D_5$ therefore has dimension $114 + 114 - 106 = 122$.

For each of all 87 observers, let $\Lambda_j = \ker(\overline{O}_j|_S)$ denote its local kernel in this sum. The desired pair core is

$$P = \bigcap_{i < j} (\Lambda_i + \Lambda_j). \quad (9)$$

Computing this intersection directly is possible but gives an awkward certificate. The annihilator identity below turns it into a sum of row-space intersections.

Theorem 4 (Dual pair-core certificate). *Let S be finite-dimensional, $\Lambda_j \leq S$, and $\mathcal{R}_j = \Lambda_j^\perp \leq S^*$. Then*

$$\left(\bigcap_{i < j} (\Lambda_i + \Lambda_j) \right)^\perp = \sum_{i < j} (\mathcal{R}_i \cap \mathcal{R}_j). \quad (10)$$

Consequently, if $C \leq \Lambda_i + \Lambda_j$ for every pair and the supplied covectors on the right-hand side span $C^\perp$, then $C = P$.

On the target, 42 independent dual witnesses prove that the intersection has codimension 42 in the 122-space and therefore dimension 80. This certificate never assumes that pairwise ranks predict a global intersection. The identity, its certificate form, and the relevant finite-dimensional lattice lemmas are Lean-checked.

3.5 Step 10: the direct 64 + 16 carrier

The intersection of all 87 local kernels Λ_j inside P is a common invisible space N_0 of dimension 64. This is exactly the retained space T from Section 3.2: T lies in both candidate spaces and every local kernel, hence in P and N_0 ; the measured equality $\dim N_0 = \dim T = 64$ gives $N_0 = T$. Thus these two structural occurrences of 64 are connected, whereas the Krylov length margin counts extra terms. Passing to the quotient gives

$$C = P/N_0, \quad \dim C = 80 - 64 = 16. \quad (11)$$

Each observer cuts out an 8-dimensional subspace $L_j \leq C$, and every one of the 3,741 pairs satisfies $\dim(L_i + L_j) = 16$. Thus distinct local spaces are complementary. This is the decisive direct-carrier event: the useful object is already a

16-dimensional binary space with 8-dimensional local lines, so the attack does not need to enumerate a family of codimension-one lifts. In the notation of the alternative unprojection route (Section E), the exponential factor 2^e is bypassed rather than merely bounded.

The dimension profile alone is not accepted as proof of hidden Goppa geometry. It is an admission test for the next deterministic calculation. A false 16-space can survive only until graph-pencil replay or, at the latest, the complete public-key verification of Step 13.

3.6 Step 11: recognizing a projective line

Choose two complementary local spaces and write $C = L_0 \oplus L_\infty$. Every other local space complementary to L_∞ is the graph of a binary linear map $M_j : L_0 \rightarrow L_\infty$. A third local space identifies the two halves and is normalized as L_1 . If the 87 subspaces arise from hidden locators in $\mathbb{F}_{256}$, the maps M_j lie in one conjugate copy of the field inside the algebra of 8×8 binary matrices. In plain terms, addition and composition of these matrices reproduce addition and multiplication in $\mathbb{F}_{256}$.

The graph-pencil routine fixes public labels 66, 114, 145 as the projective points $0, \infty, 1$, recognizes the common field action, and solves for the remaining points. The eigenvalues encode fourth powers of the normalized locator cross-ratios; this is the fourth-power relation checked in replay. Changing the binary basis of $\mathbb{F}_{256}$ by Frobenius $x \mapsto x^{2^r}$ gives eight equally valid branches. On TII-254, all eight branches contain 87 distinct projective points, reproduce every local kernel and fourth-power relation, and form one Frobenius orbit. No branch is selected by a secret-dependent score.

3.7 Steps 12–13: completion and definitive verification

The graph pencil recovers 87 of the 223 support locators only up to a projective change of coordinates. The public circuit consists of 47 columns of rank 46, with exactly one label outside those 87. Choosing a pole and the locator value of that one outside label (the *circuit root*) supplies an 88th locator. The projective line over $\mathbb{F}_{256}$ has 257 points, so there are 170 unused pole choices followed by 169 root choices, for a conservative inventory of

$$170 \cdot 169 = 28,730 \tag{12}$$

ordered trials per admitted branch. Characteristic-two derivative identities turn each trial into a numerator whose degree-12 factors are candidate Goppa polynomials. Public row-space tests, a semantic codeword-support test, and the Kirshanova–May completion method [8] then recover the remaining support positions.

Theorem 5 (Equivalent-key verification). *Let (α, g) have distinct support entries, let g be monic and irreducible with $\deg g = t$, and suppose $g(\alpha_i) \neq 0$.*

Table 1. Certified dimensions in the successful TII-254 route.

Object or certificate	Dimension or rank
Literal degree-six source	20,271,300
Certified nuisance space L_*	4,617,031
Coordinate complement	15,654,269
Anchor-conditioned square operator B_a	15,527,170
Finite-Hankel rank, each anchor	15,527,049
Complete compressed kernels $\widehat{K}_2, \widehat{K}_5$	121, 121
Replay-certified physical kernels K_2, K_5	121, 121
Intersection $K_2 \cap K_5$	106
Observer-compatible candidates D_2, D_5	114, 114
Physical candidate sum $D_2 + D_5$	122
Dual pair-core witness	42
Certified pair core P	80
Common invisible space N_0	64
Locator carrier P/N_0	16
Each of 87 local cuts in the carrier	8

Form the $mt \times n$ binary matrix obtained by expanding the rows $(\alpha_i^k/g(\alpha_i))_{i=1}^n$ for $0 \leq k < t$ in any binary basis of $\mathbb{F}_{2^m}$. If its row space equals the row space of the public parity-check matrix H , then (α, g) defines exactly the public binary Goppa code and is an equivalent decoding key.

The condition is basis-independent because changing the field basis only applies an invertible binary row transformation. It is also stronger than checking a subset of public columns. Our standalone Sage verifier imports no recovery code and independently checks all 223 columns, matrix rank 96, polynomial irreducibility and nonvanishing, support distinctness, and exact row-space equality. A fresh deterministic replay of the recovery implementation reproduced the locator branch, pole, outside point, polynomial, support, and screening transcript.

4 The TII-254 computation

4.1 Exact dimension transcript

Table 1 records the dimension changes that carry the computation from the literal degree-six system to the locator carrier. These are exact ranks over $\mathbb{F}_2$, not expected values.

The two independent equalities $15,527,170 - 15,527,049 = 121$ are especially important. They change the interpretation of reconstruction: the recovered panels are complete kernels under an exact upper bound, not partial samples with a guessed missing dimension. The label-5 panel contributes only 15 new dimensions modulo K_2 , exactly matching the relative-completion requirement.

Table 2. Per-anchor resources for the main computation stages. Elapsed computing times use hh:mm:ss and exclude queueing. CPU counts are logical threads.

Compute Step	Label 2	Label 5
Krylov Sequence	13:36:17, 1 $\times$ GH200	13:36:10, 1 $\times$ GH200
Polynomial-matrix basis	05:05:53, 12 $\times$ CPU	00:49:41, 64 $\times$ CPU
Reconstruction	Full time unavailable 8 $\times$ GH200	01:55:23 1, then 7 GH200s

4.2 Large linear algebra

For each anchor, we generated a width-512, 60,718-term Krylov sequence on a GPU, computed a polynomial-matrix basis on CPUs, and reconstructed the kernel vectors on GPUs. Table 2 lists elapsed computing times and hardware. GPU-hours add the time used by all GPUs, including those running in parallel.

The two sequences account for 27.2 GPU-hours. For label 5, reconstruction used one GPU for about 58 minutes, then seven GPUs in parallel for up to another 58 minutes, giving 01:55:23 of elapsed computing time. Label 2 used eight GPUs in parallel, but its complete reconstruction time was not recorded. The basis times exclude validation and other job overhead. These measurements therefore do not give a complete recovery time or total GPU-hour cost.

Each reconstruction task requested 64 GiB (2^{39} bits) of host memory. On the label-2 run, the scheduler on CSC’s Roihu supercomputer in Finland^{***} raised this request to its partition minimum of 212 GiB per task; this scheduler allocation must not be confused with measured use. A reconstruction task used about $2^{39.19}$ bits of GPU memory. The label-5 polynomial-basis task requested 128 GiB (2^{40} bits), while the measured label-2 polynomial-basis process-family peak was about $2^{37.05}$ bits. These figures describe the target implementation, not an asymptotic lower bound; the algorithm can trade wall time against the number of parallel tasks and block width.

The overall effort, including development, earlier experiments, verification, and packaging, lasted about eight days and fifteen hours. This is calendar time, not the running time of the final calculation. The artifact provides the public matrix, recovered key, and compact pair-core checkpoint for verification and downstream recovery without repeating the large computations.

4.3 Locator completion and recovered key

The certified 16-space produced eight Frobenius-equivalent branches of 87 locators. On the successful branch, the finisher used the 28,730 pole/root inventory as its bounded search domain but stopped at the first verified key. It therefore factored only 1,727 derivative numerators, obtaining 152 factor candidates. All 152 passed the restricted public row-space test; the semantic screens rejected 151, leaving one survivor. One Kirshanova–May completion attempt extended

^{***} Roihu system documentation: <https://docs.csc.fi/computing/systems-roihu/>.

the 88 known support positions to 97 after nine set-intersection steps. These known columns had full binary rank 96, fixing the row-basis transformation between the public matrix and the canonical Goppa check matrix. Applying that transformation to the remaining public columns and matching each to its unique canonical Goppa column recovered all 223 support positions. The measured finisher wall time was 37.99 seconds.

The recovered monic irreducible degree-12 polynomial and its finite-field coefficient encoding are given in Section A. Together with the 223-element support stored in the artifact, it reproduces the authenticated public row space.

We again stress that the result is an equivalent binary-Goppa decoding key, not a claim about the challenge author’s original representation.

4.4 What remains heuristic, and what does not

The choice of degree, public chart, anchors, block width, and Krylov projections was guided by theory and experiments. A different instance can cause a projected sequence to miss kernel directions, can yield conditioned kernels with an unhelpful intersection, or can fail to expose a direct 16-dimensional carrier. The method therefore does not prove that every parameter-compatible Goppa key is recovered with the same cost.

None of these qualifications weaken the positive TII-254 result. Completeness of both target kernels follows from Hankel upper bounds and replayed lower bounds; relative completion is an equality; the 80-space has a dual certificate; the $64 + 16$ quotient and all local ranks are exact; every locator branch is replayed; and the final key is checked against every public column. The remaining probabilistic choices affect the chance and cost of reaching a certificate, not the soundness of a certificate once reached.

5 Conclusion

TII-254 is solved by a two-anchor, observer-amortized Holdout/Hermite relation attack. The essential advance was not simply a larger block-Wiedemann run. It was a chain of exact reductions that made the output interpretable: coordinate deflation removed 4.6 million certified nuisance directions, finite-Hankel ranks proved that two 121-dimensional kernels were complete, relative completion reduced the second kernel’s new information to 15 dimensions, and a dual pair-core certificate exposed a direct $64 + 16$ decomposition. The 16-dimensional quotient then revealed the $\mathbb{F}_{256}$ projective geometry needed by a short deterministic key finisher.

The result also gives a useful design principle for future algebraic attacks. Exact, separately replayable rank statements should surround large projected linear algebra. This turns an experiment that “found some relations” into an algorithm with explicit success gates and makes the boundary between a finite miss and a genuine structural obstruction visible.

Availability. We provide a self-contained artifact containing the public input and recovered key, a compact pair-core checkpoint, Python/Sage code for downstream recovery and independent key verification, CUDA and CPU sources for the large linear-algebra stages, and Lean proofs of the reusable theorems. The large production inputs and relation panels needed to reconstruct and certify the pair core are not included; the compact computation starts from that core. Repository: <https://github.com/mjosaarinen/tii254-artifact>

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Use of Artificial Intelligence

These results originate from a systematic effort to implement and improve cryptanalytic attacks against Post-Quantum Cryptography algorithms with AI help. We used OpenAI (GPT-6 Astra, GPT-5.6 Sol) and Anthropic (Fable 5.1, Opus 5) models to develop theory, write code, run experiments, analyze results, and draft the final presentation – all under close human supervision.

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A Recovered TII-254 Goppa polynomial

The recovered polynomial is defined over

$$F = \mathbb{F}_2[Z]/(Z^8 + Z^4 + Z^3 + Z^2 + 1). \quad (13)$$

Write θ for the residue class of Z . This is the artifact’s polynomial-basis representation with binary modulus `0x11d`. To specify coefficients compactly, for a byte $c \in \{0, \dots, 255\}$ define

$$[c]_\theta = \sum_{j=0}^7 c_j \theta^j, \quad c = \sum_{j=0}^7 c_j 2^j, \quad c_j \in \{0, 1\}. \quad (14)$$

Thus, for example, $[128]_\theta = \theta^7$ and $[8]_\theta = \theta^3$; the decimal byte labels specify field elements through (14). The monic irreducible Goppa polynomial returned by the successful computation is

$$\begin{aligned} g(X) = & X^{12} + [128]_\theta X^{11} + [183]_\theta X^{10} + [8]_\theta X^9 \\ & + [96]_\theta X^8 + [76]_\theta X^7 + [19]_\theta X^6 + [92]_\theta X^5 \\ & + [71]_\theta X^4 + [177]_\theta X^3 + [230]_\theta X^2 \\ & + [244]_\theta X + [243]_\theta. \end{aligned} \quad (15)$$

Equivalently, its encoded coefficients in ascending order, from the constant term to the coefficient of X^{12} , are

$$(243, 244, 230, 177, 71, 92, 19, 76, 96, 8, 183, 128, 1). \quad (16)$$

The polynomial coefficients and ordered support are stored in the artifact’s `data/recovered_key.json`. The independent verifier described in Section D checks polynomial irreducibility, nonvanishing on that support, and equality with the public row space. This polynomial and support give an equivalent representation of the challenge code.

B Proofs of the reusable linear-algebra gates

B.1 Coordinate deflation

Let $v \in V$. By surjectivity of the coordinate projection $L \rightarrow \mathbb{F}_2^P$, there is a unique $\ell \in L$ having the same coordinates as v on P . Then $s = v - \ell$ lies in S_P ,

so $V = L + S_P$. The intersection is zero because a vector in $L \cap S_P$ projects to zero and the projection is injective on L . Since $T(\ell) = 0$, we have $T(v) = T(s)$ and hence $T(V) = T(S_P)$. Finally, the quotient map sends $\ker(T|_{S_P})$ bijectively to $\ker T/L$ under the direct-sum decomposition.

For TII-254, the required information set is block triangular. Its ordinary degree-four block has 4,575,736 pivots, its secondary degree-three block has 32,943, and its degree-two block has 8,352, totalling 4,617,031. The literal generator family has the same exact rank and every generator replays to zero.

B.2 The shifted-Hankel bound

Define block maps $\mathcal{Y}(x_0, \dots, x_s) = \sum_j B^j Y x_j$ and $\mathcal{U}(v) = (U^\top B^i v)_i$. The shifted Hankel matrix factors as $\mathcal{H} = \mathcal{U} B \mathcal{Y}$, so $\text{rank } \mathcal{H} \leq \text{rank } B$. Rank-nullity gives $\dim \ker B = N - \text{rank } B \leq N - \text{rank } \mathcal{H}$. If $Q \leq \ker B$ reaches the upper bound, both spaces have the same dimension and are equal.

This argument does not assert that a finite Krylov projection always reaches $\text{rank } B$. It says that the observed Hankel rank is an unconditional lower bound on $\text{rank } B$. Equality with a separately replayed kernel lower bound closes the gap on the target.

B.3 Relative completion

Because $P_5 \leq K_5$, dimension modularity gives $\dim(K_2 + P_5) - \dim K_2 = \dim P_5 - \dim(P_5 \cap K_2) \leq \dim K_5 - \dim(K_5 \cap K_2) = \dim K_5 - \dim I \leq \kappa_5 - \dim I$. Equality at the two ends forces $\dim K_5 = \kappa_5$ and equality of the intermediate quotient images. Hence every coset of K_5/I has a representative in P_5 , which is exactly $K_5 = I + P_5$.

B.4 The dual pair-core identity

In finite dimension, $(\bigcap_a V_a)^\perp = \sum_a V_a^\perp$ and $(U + V)^\perp = U^\perp \cap V^\perp$. Applying these two identities to $V_{ij} = A_i + A_j$ proves Eq. (10). For the certificate form, containment $C \leq P$ gives $P^\perp \leq C^\perp$. If explicit covectors known to lie in $P^\perp$ span $C^\perp$, then $C^\perp \leq P^\perp$ as well; equality of annihilators gives $C = P$.

C The certified nuisance space

The deflation compiler uses square-free monomial spaces $\mathcal{A}_r$ with bases indexed by r -subsets of the 50 affine coordinates. It constructs the following independent families in $\ker \mathcal{E}$.

The independence proof orders pivots by degree and endpoint. Within an ordinary block, an endpoint-triangular minor separates the pointwise arms and pair cycles. The three cross-block families then occupy fresh rows in the secondary and degree-two stages. This construction proves the displayed dimension

Table 3. Independent components of L_* .

Family	Dimension
Ordinary pointwise family, first arm	106,488
Ordinary pointwise family, second arm	87
Secondary pointwise family	7,350
Ordinary pair cycles modulo pointwise families	4,469,248
Cross-block family A	8,265
Cross-block family B	25,578
Cross-block family C	15
Total	4,617,031

and the coordinate-complement theorem; it does not claim that no further multi-block identities exist. That stronger completeness question is unnecessary for the successful route because any additional relations remain in the reduced kernel and are captured by the later exact rank computation.

D Formal and executable artifact

The repository URL is given in the availability statement. All paths below are relative to the artifact root. It includes the preprint as `tii254.pdf` and separates compact executable recovery, source code for the large stages, and formal linear-algebra proofs.

Compact recovery and verification. The supplied public matrix and recovered key are in `data/public_key.txt` and `data/recovered_key.json`. From the artifact root, `sage verify_key.sage` reconstructs $\mathbb{F}_{256}$ using modulus `0x11d`, reads the 223 support elements and thirteen polynomial coefficients, constructs the full Goppa parity-check matrix, and compares its exact row space with the public one. It checks support distinctness, polynomial irreducibility, degree, non-vanishing, and rank. Acceptance rests on this exact algebraic comparison.

The compact checkpoint `data/pair_core.json` contains the 80-dimensional core and the common candidate space in 122 physical coordinates, together with the local observer kernels expressed in core coordinates. The Python code recomputes the $64 + 16$ quotient and its local ranks, then recognizes the locator geometry; the Sage code completes and verifies a key. The checkpoint does not contain the large relation panels needed to reconstruct the complete anchor kernels or independently re-establish the original dual pair-core certificate.

CUDA and CPU sources. The `cuda/` directory contains the production-derived matrix-free relation operator, anchor codecs, projected Krylov workers, and parallel kernel-reconstruction workers for the supplier stages in Section 3.

The CADO-NFS adapters `scripts/sequence_to_cado.c`, `scripts/make_cado_aux.py`, and `scripts/extract_relations.cpp` connect the projected sequence to an external CADO-NFS polynomial-basis computation and extract

the reconstruction relations. The `rust/` workspace contains the CPU implementation of the relation operator $\mathcal{E}$, called `Tensor-Bier` in the code, and the independent original-coordinate replay code for $\mathcal{E}(JQ) = 0$.

These native sources are included for inspection and reuse, not as a turnkey rerun of the complete target calculation. The multi-gigabyte operator/codec inputs, Krylov checkpoints, and full relation panels are omitted. The interfaces and build requirements are documented in `docs/FORMATS.md`, `cuda/README.md`, and the root `README`; compact key verification does not require a GPU or a CADO-NFS build.

Formal proofs. The project under `lean/` pins Lean 4.33.0 and its Mathlib dependencies. From that directory, `lake exe cache get` obtains the dependency cache and `lake build` checks the project. Table 4 maps the paper’s gates to modules under `lean/ProjectedKernelRigidity/`. These files prove abstract linear-algebra statements, not the challenge-specific million-dimensional ranks. Replaying those large computations requires the omitted production inputs; neither the Lean build nor the compact recovery claims to do so.

Table 4. Lean-checked reusable results.

Paper gate	Lean module
Coordinate deflation	<code>CoordinateDeflation</code>
Shifted-Hankel rank and kernel completeness	<code>ZeroPrimaryReachability</code>
Relative second-anchor completion	<code>RelativeAnchor</code>
Dual pair-core certificate	<code>TwoAnchorPairCore</code>
Toeplitz transposition and retry bounds	<code>ToeplitzStart</code>
Observer-completeness compiler interfaces	<code>ObserverCompletenessFinite</code>

From `lean/`, the command `lake env lean AxiomAudit.lean` prints the axioms of the headline formal statements. The supplied `python3 audit.py` check rejects `sorry`, `admit`, and `native.decide` in the proof modules and permits only the standard axioms `propext`, `Classical.choice`, and `Quot.sound` in the audited theorems.

E Evidence boundary and alternative route

Before the direct $64 + 16$ target core was observed, the contingency plan treated an 80-dimensional pair core as a codimension-one projection of a larger conic carrier. Lifting such a projection can leave e binary degrees of freedom and therefore a naive 2^e enumeration. The target result made this contingency unnecessary: quotienting by the certified common 64-space immediately produced the required 16-space. Thus no unprojection conjecture or exponential enumeration enters the TII-254 success claim.

The Toeplitz right-start analysis addresses a different potential obstruction. For a fixed rank- g quotient and a uniformly sampled width- b Toeplitz panel over $\mathbb{F}_q$, the probability that one panel is invisible is bounded by q^{g-b} ; a union bound and a geometric retry calculation require no independence among quotient directions. This theorem supports preregistered retry policies, but this paper does not use it to assign a retrospective success probability to the fixed challenge. Instead, Theorems 2 and 3 certify the two kernels as complete.

Finally, the graph-pencil and finisher stages rely on literal replay rather than a distributional claim. A candidate geometry that happens to have the right dimensions still has to reproduce all observer kernels, all locator power relations, the public circuit, the support-completion transcript, and the entire 96×223 public row space. Consequently, the artifact’s positive verification is definitive even though the probability that another instance reaches the same carrier profile remains an open ensemble question.